Extended Derivations and Results for "What Does a Meta-Learned Horizon Weighting Learn? A Shrinkage View of Quadratic Direct Forecasting"

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This document accompanies the code package. It is not part of the two supplementary pages of the letter. It gives the derivations behind Propositions 1–3 and the linear-forecaster argument in more detail, the numerical checks of the algebra, the replay of the released meta phase, the meta step budgets of the released scripts, all seventeen settings of the earlier protocols, the full results of protocol v8, and the frozen verdicts of all protocols. Equation numbers without the prefix E refer to the letter. All statements concern the released QDF code at commit `eb0693a9` with its released flags: first-order meta-gradients, plain SGD as meta optimizer, MSE as auxiliary loss, no regularizer on Σ, and no log-determinant term.

I. Setting

For one sample and one channel, $e = \hat{y} - y \in \mathbb{R}^T$ is the residual over the horizon. The code parametrizes $L = \mathrm{tril}(P) + \varepsilon I$ with $\varepsilon = 10^{-6}$ and $P_0 = I$, and sets $\Sigma = LL^\top$. The diagonal of P is removed and re-added before ε is added, so the Jacobian of L with respect to the lower triangle of P is the identity. The strictly upper triangle of P never receives a gradient. For residual rows $\{e_n\}_{n=1}^N$ with uncentred second moment $\hat{S} = \frac{1}{N}\sum_n e_n e_n^\top$, the loss is

$$J(L;\hat{S}) = \frac{1}{NT}\sum_{n=1}^{N} \|L^{-1}e_n\|_2^2 = \frac{1}{T}\,\mathrm{tr}\left(L^{-1}\hat{S}L^{-\top}\right) = \frac{1}{T}\,\mathrm{tr}\left(\Sigma^{-1}\hat{S}\right). \tag{E1}$$

The factor $1/T$ comes from averaging over all NT entries. In meta step t, each of M tasks adapts a copy of the forecaster from the initial weights θ_0 by K clipped SGD steps on (E1), evaluates (E1) on a query minibatch, and averages over tasks. With the first-order option the adapted weights carry no autograd history, so the query second moment $\hat{S}_t$ is a constant for the gradient with respect to P. The forecaster is never stepped during the meta phase.

II. Proof of Proposition 1

Gradient. Using $d(L^{-1}) = -L^{-1}\,dL\,L^{-1}$ and the cyclic property of the trace,

$$dJ = \frac{1}{T}\,\mathrm{tr}\left(d(L^{-1})\hat{S}L^{-\top} + L^{-1}\hat{S}\,d(L^{-\top})\right) = -\frac{2}{T}\,\mathrm{tr}\left(L^{-1}\,dL\,L^{-1}\hat{S}L^{-\top}\right). \tag{E2}$$

Hence $\partial J/\partial L = -\frac{2}{T}L^{-\top}L^{-1}\hat{S}L^{-\top} = -\frac{2}{T}\Sigma^{-1}\hat{S}L^{-\top}$ for an unconstrained L. The chain rule through $P \mapsto \mathrm{tril}(P) + \varepsilon I$ applies the projection onto lower-triangular matrices, which gives (2). At $P_0 = I$, $L_0 = (1+\varepsilon)I$ and the gradient is $-\frac{2}{T}(1+\varepsilon)^{-3}\,\mathrm{tril}(\hat{S})$, which equals $-\frac{2}{T}\,\mathrm{tril}(\hat{S})$ up to a relative error of 3ε.

Homogeneity and the radial component. From (E1), $J(cL) = c^{-2}J(L)$ for $c > 0$. Euler's identity for homogeneous functions gives $\langle L, \partial J/\partial L\rangle_F = -2J$. Because L is lower triangular, $\langle L, \mathrm{tril}(G)\rangle_F = \langle L, G\rangle_F$ for every G, so $\langle L, \nabla_P J\rangle_F = -2J < 0$ whenever $\hat{S} \neq 0$. An SGD step $P \leftarrow P - \eta\nabla_P J$ therefore always has a positive component along L.

No stationary point. Let $C = L^{-1}\hat{S}L^{-\top} \succeq 0$, so that $\partial J/\partial L = -\frac{2}{T}L^{-\top}C$. Suppose $\mathrm{tril}(L^{-\top}C) = 0$. Then $V = L^{-\top}C$ is strictly upper triangular, and $C = L^\top V$ is the product of an upper-triangular and a strictly upper-triangular matrix, hence strictly upper triangular. A symmetric strictly upper-triangular matrix is zero, so $C = 0$ and $\hat{S} = 0$.

Trace increment. The update $P \leftarrow P - \eta G$ with the lower-triangular $G = \nabla_P J$ changes $L = \mathrm{tril}(P) + \varepsilon I$ to $L - \eta G$, including the ε term. Since $\mathrm{tr}\,\Sigma = \|L\|_F^2$,

$$\|L - \eta G\|_F^2 - \|L\|_F^2 = -2\eta\langle L, G\rangle_F + \eta^2\|G\|_F^2 = 4\eta J + \eta^2\|G\|_F^2 > 0. \tag{E3}$$

The trace of Σ therefore grows at every meta step, for any step size and any query moment $\hat{S}_t \neq 0$. When the meta loss is averaged over tasks, J and $\hat{S}_t$ are the task averages. □

III. Proof of Proposition 2 and the role of Adam

One step. One SGD step of size η from $P_0 = I$ gives, up to $O(\varepsilon)$, $L_1 = I + \beta U$ with $U = \operatorname{tril}(\hat{S}_0)$ and $\beta = 2\eta/T$. Expanding,

$$\Sigma_1 = (I + \beta U)(I + \beta U)^\top = I + \beta\big(\hat{S}_0 + \operatorname{Diag}\hat{S}_0\big) + \beta^2 U U^\top, \tag{E4}$$

because $U + U^\top = \hat{S}_0 + \operatorname{Diag}\hat{S}_0$. The diagonal of $\hat{S}_0$ therefore enters Σ with twice the weight of the off-diagonal entries.

Many steps. Write $L_t = I + \Delta_t$ and $\tau_t = \frac{2}{T}\sum_{s<t}\eta_s$. Expanding (2) around $L = I$,

$$\Sigma^{-1}\hat{S}L^{-\top} = \hat{S} - (\Delta + \Delta^\top)\hat{S} - \hat{S}\Delta^\top + O(\|\Delta\|^2\|\hat{S}\|), \tag{E5}$$

so $\Delta_{t+1} - \Delta_t = \frac{2\eta_t}{T}\operatorname{tril}(\hat{S}_t) + O\big(\frac{\eta_t}{T}\|\Delta_t\|\|\hat{S}_t\|\big)$. Since $\|\Delta_t\| = O(\tau_t\|\hat{S}\|)$, summing over t gives $\Delta = \alpha\operatorname{tril}(\bar{S}) + O(\alpha^2 s^2)$ with $\alpha = 2\eta_{\text{tot}}/T$, $\bar{S} = \sum_t \eta_t\hat{S}_t/\eta_{\text{tot}}$ and $s = \max_t\|\hat{S}_t\|_2$. Inserting this into $\Sigma = (I + \Delta)(I + \Delta)^\top$ as in (E4) gives (3). The approximation is first order in αs. Section VII shows that the released budgets are far outside this regime.

Second-order term. For constant $\hat{S}_t \equiv S$ and gradient flow in τ, with $U = \operatorname{tril}(S)$ and $D = \operatorname{Diag} S$,

$$\Sigma(\tau) = I + \tau(S + D) + \tau^2 R + O(\tau^3), \quad R = UU^\top - \tfrac{1}{2}\big[\operatorname{tril}(M) + \operatorname{tril}(M)^\top\big], \quad M = (S + D)S + SU^\top. \tag{E6}$$

The negative part of R bends the path away from the straight line $I + \tau(S + D)$. Discrete SGD adds a term of order $\max_t(2\eta_t/T)\,\tau\|S\|^2$.

Lemma E1 (Scale invariance of the trained forecaster)**.** *In the meta-test phase the forecaster is trained with Adam on* (E1) *with Σ fixed, and early stopping and the learning-rate schedule use the validation MSE. Replacing Σ by $c\Sigma$ with $c > 0$ multiplies the loss and every gradient by $1/c$. The Adam update $\hat{m}/(\sqrt{\hat{v}} + \epsilon_{\text{A}})$ is unchanged when $\epsilon_{\text{A}} = 0$, and changes only through ϵ_{A} otherwise.*

Proof. Both moment estimates scale as $\hat{m} \mapsto \hat{m}/c$ and $\hat{v} \mapsto \hat{v}/c^2$, so their ratio is invariant when $\epsilon_{\text{A}} = 0$. The validation MSE does not involve Σ. □

Only the shape $\Sigma/(\operatorname{tr}\Sigma/T)$ therefore affects the trained forecaster. Writing $\bar{S} = \frac{\operatorname{tr}\bar{S}}{T}\bar{S}_{\text{n}}$ with $\operatorname{tr}\bar{S}_{\text{n}} = T$ turns (3) into (4), with $\rho = \alpha\operatorname{tr}\bar{S}/T = 2\eta_{\text{tot}}\operatorname{tr}\bar{S}/T^2$. Up to scale, $I + \rho\bar{S}_{\text{n}} = (1 + \rho)[(1 - \lambda)I + \lambda\bar{S}_{\text{n}}]$ with $\lambda = \rho/(1 + \rho)$, which is the linear shrinkage form of Ledoit and Wolf. The analogy is structural: here λ is set by the step budget, not by an estimation-error criterion.

Adam as meta optimizer (not the released setting). With bias correction, the first Adam step is $-\eta\operatorname{sign}(g)$ entrywise, so $L_1 = I + \eta\operatorname{sign}(\operatorname{tril}\hat{S}_0)$ and the magnitudes of $\hat{S}_0$ are lost. Proposition 2 does not hold in that case.

IV. Proof of Proposition 3 and the diagonal flow

Inequality. This is the trace-constrained minimization in Lemma 15 of Duchi, Hazan and Singer (J. Mach. Learn. Res., 2011). A direct proof: for $\hat{S} \succ 0$ and $\Sigma \succ 0$ let $A = \Sigma^{-1/2}\hat{S}^{1/2}$ and $B = \Sigma^{1/2}$. Then $\langle A, B\rangle_F = \operatorname{tr}(\hat{S}^{1/2}\Sigma^{-1/2}\Sigma^{1/2}) = \operatorname{tr}\hat{S}^{1/2}$, $\|A\|_F^2 = \operatorname{tr}(\Sigma^{-1}\hat{S})$ and $\|B\|_F^2 = \operatorname{tr}\Sigma$. The Cauchy–Schwarz inequality gives (5), with equality if and only if $A = cB$ for some $c > 0$, that is, $\hat{S}^{1/2} = c\Sigma$. Every $\Sigma \succ 0$ has a Cholesky factor, so the minimum over the triangular parametrization is attained at $\Sigma \propto \hat{S}^{1/2}$.

Monotonicity. Consider the flow $\dot{L} = -G$ with $G = \nabla_P J$ on invertible lower-triangular matrices, with constant $\hat{S} \succ 0$, and let $F(L) = J(L)\,\|L\|_F^2 = J\operatorname{tr}\Sigma$. Restricted to lower-triangular directions, $\nabla F = \|L\|_F^2 G + 2JL$, so by Proposition 1

$$\dot{F} = -\|L\|_F^2\|G\|_F^2 - 2J\langle L, G\rangle_F = 4J^2 - \|L\|_F^2\|G\|_F^2 \le 0, \tag{E7}$$

because $2J = |\langle L, G\rangle_F| \le \|L\|_F\|G\|_F$. Equality holds if and only if $G = \mu L$ for some μ, and $\mu < 0$ because $\langle L, G\rangle_F = -2J < 0$.

Equality set. Write $G = -\frac{2}{T}\operatorname{tril}(L^{-\top}C)$ with $C = L^{-1}\hat{S}L^{-\top}$, and suppose $\operatorname{tril}(L^{-\top}C) = \nu L$ with $\nu > 0$. Then $L^{-\top}C = \nu L + V$ with V strictly upper triangular, so $C - \nu L^\top L = L^\top V$ is symmetric and strictly upper triangular, hence zero. Thus $L^{-1}\hat{S}L^{-\top} = \nu L^\top L$, that is, $\hat{S} = \nu\Sigma^2$ and $\Sigma = \hat{S}^{1/2}/\sqrt{\nu}$. Conversely, $\Sigma = c\hat{S}^{1/2}$ gives $L^{-\top}C = c^{-2}L$, which is lower triangular. So $\dot{F} = 0$ exactly on the ray $\Sigma \propto \hat{S}^{1/2}$, where F attains its lower bound $(\operatorname{tr}\hat{S}^{1/2})^2/T$.

Convergence of the shape. Since $J(cL) = c^{-2}J(L)$, the gradient is homogeneous of degree -3, and the normalized factor $\tilde{L} = L/\|L\|_F$ obeys $d\tilde{L}/dt = -\|L\|_F^{-4}\big(G(\tilde{L}) - \langle \tilde{L}, G(\tilde{L})\rangle_F \tilde{L}\big)$. In the time $u = \int_0^t \|L\|_F^{-4} dt'$, $\tilde{L}$ follows an autonomous flow on the unit sphere of lower-triangular matrices along which $F(\tilde{L}) = F(L)$ is non-increasing. The new time diverges: by (E3) in continuous form, $d\|L\|_F^2/dt = 4J = 4F/\|L\|_F^2 \le 4F(0)/\|L\|_F^2$, hence $\|L\|_F^4 \le \|L(0)\|_F^4 + 8F(0)t$ and $u(t) \ge \int_0^t (\|L(0)\|_F^4 + 8F(0)t')^{-1} dt' \to \infty$. On the sphere, $F(\tilde{L}) \ge \lambda_{\min}(\hat{S})\|\tilde{L}^{-1}\|_F^2/T$ diverges as $\tilde{L}$ approaches a singular matrix, so the sublevel set $\{F \le F(0)\}$ is compact and contains no singular matrix. In particular the diagonal of L, positive at $L(0) = I$, stays positive. By LaSalle's invariance principle the trajectory approaches the largest invariant subset of $\{\dot{F} = 0\}$ in this set, which consists of the single normalized Cholesky factor of $\hat{S}^{1/2}$ with positive diagonal. Hence $\Sigma/\operatorname{tr}\Sigma = \tilde{L}\tilde{L}^\top \to \hat{S}^{1/2}/\operatorname{tr}\hat{S}^{1/2}$. Because u grows only logarithmically in t when $\|L\|_F^4$ grows linearly, the approach is slow, and a finite step budget stops at an intermediate point of the path. □

Contrast with the Gaussian likelihood. Adding $\log\det\Sigma$ to $\operatorname{tr}(\Sigma^{-1}\hat{S})$ gives the negative Gaussian log-likelihood, whose stationarity condition $\Sigma^{-1} - \Sigma^{-1}\hat{S}\Sigma^{-1} = 0$ yields $\Sigma = \hat{S}$. The released objective omits this term, which is why it has no stationary point and why its scale-free version favours $\hat{S}^{1/2}$ instead of the GLS weighting $\hat{S}$.

Diagonal flow. For $\hat{S} = \operatorname{diag}(s_i)$ and diagonal $L = \operatorname{diag}(l_i)$, $J = \frac{1}{T}\sum_i s_i/l_i^2$ and the gradient flow in the cumulative step $\tau' = \sum_t \eta_t$ is

$$\frac{dl_i}{d\tau'} = \frac{2}{T}\frac{s_i}{l_i^3} \quad\Longrightarrow\quad \frac{d(l_i^4)}{d\tau'} = \frac{8s_i}{T} \quad\Longrightarrow\quad \Sigma_{ii} = l_i^2 = \Big(1 + \frac{8s_i\tau'}{T}\Big)^{1/2}. \tag{E8}$$

For small τ' this is $1 + 4s_i\tau'/T = 1 + 2\alpha s_i$, the doubled diagonal of (3). For large τ', $\Sigma \propto \hat{S}^{1/2}$, as the general result above predicts. For a general $\hat{S}$ the path has no closed form, because the triangular projection breaks rotation equivariance; Section VII computes it numerically.

V. Linear forecasters: invariance and the gradient-descent recursion

Consider $y_n = Bx_n + u_n$ with $y_n \in \mathbb{R}^T$ and a regressor $x_n \in \mathbb{R}^p$ shared by all T outputs, $X = [x_1, \dots, x_N]$, a fixed $W \succ 0$ and fixed per-sample weights $\omega_n > 0$. The normal equations of $\sum_n \omega_n (y_n - Bx_n)^\top W (y_n - Bx_n)$ are $W\sum_n \omega_n (y_n - Bx_n)x_n^\top = 0$. W is invertible and cancels, so the set of minimizers does not depend on W; when $\sum_n \omega_n x_n x_n^\top$ is invertible it is the single point $B^\star = (\sum_n \omega_n y_n x_n^\top)(\sum_n \omega_n x_n x_n^\top)^{-1}$. In Kruskal's form, the design $Z = X^\top \otimes I_T$ satisfies $(I_N \otimes W^{-1})Z = Z(I_p \otimes W^{-1})$, so its column space is invariant under the error covariance, which is the condition for generalized and ordinary least squares to coincide. Per-sample weights ω_n cover instance normalization, which rescales the whole horizon of a sample by one scalar.

For DLinear, x_n stacks the seasonal and trend parts of the input window. Both are linear in the window, so $XX^\top$ has rank at most the window length and is singular. The minimizer is then not unique, but the set of minimizers is still independent of W.

Gradient descent. With $\omega_n \equiv 1$ and any minimizer $B^\star$, the gradient of $\frac{1}{N}\sum_n (y_n - Bx_n)^\top W (y_n - Bx_n)$ is $-\frac{2}{N}W(Y - BX)X^\top = \frac{2}{N}W\Delta XX^\top$ with $Y = [y_1, \dots, y_N]$ and $\Delta = B - B^\star$, because $(Y - B^\star X)X^\top = 0$. One step with size η gives

$$\operatorname{vec}\Delta_{k+1} = \Big(I - \tfrac{2\eta}{N}\, XX^\top \otimes W\Big)\operatorname{vec}\Delta_k. \tag{E9}$$

The component of Δ in the null space of $XX^\top$ (acting from the right) never moves, and for small enough η the remaining component contracts to zero. The limit is therefore $B^\star + \Delta_0\Pi$, with Π the projector onto that null space, for every $W \succ 0$; W only changes the contraction rates along output directions. A weighting can therefore change a linear forecaster only through a finite or adaptive optimization path. Early stopping and Adam, which rescales coordinates, both act on that path, and this recursion does not describe them. In our DLinear runs such path effects were not small: a fixed Σ_a with $a = 1$ changed the test MSE by up to -3.1%.

VI. Numerical checks of the algebra

A separate script re-implements the loss, the parametrization of L and the first-order inner loop of the released code on a toy problem ($T = 8$, float64, CPU, single thread, 1.3 s). A second script (`toy_checks_flow.py`, 26 s) checks (E3) and the flow statements of Proposition 3, approximating the flow by 2×10^5 descent steps of size 10^{-2} from $L = I$. Neither script imports the QDF code, and neither says anything about forecasting accuracy. Table E1 lists the checks; all passed.

TABLE E1
NUMERICAL CHECKS OF THE DERIVATIONS (TOY PROBLEM)

Check	Statement	Error
A0, A2	Autograd gradient equals (2) at $L = I$ and at a generic L	4.4×10^{-16}
A1	Gradient at $L = I$ equals $-\frac{2}{T}\,\mathrm{tril}(\hat{S})$ (relative; differs by 3ε)	3.0×10^{-6}
G	$\langle L, \nabla_P J\rangle_F + 2J = 0$	0
B1	One-step Σ_1 equals (E4) (relative)	1.9×10^{-6}
C1	First-order meta-gradient equals (2) at the adapted residual	4.4×10^{-16}
D	First Adam step equals $-\eta\,\mathrm{sign}(g)$ on the lower triangle	6.6×10^{-8}
E	W-weighted least squares equals ordinary least squares	3.0×10^{-15}
F	Diagonal flow matches $\Sigma_{ii} = (1 + 8 s_i \tau'/T)^{1/2}$ (relative)	5.4×10^{-4}
I	Error ratio of (E6) between $\tau = 0.1$ and 0.05 is near 8 ($O(\tau^3)$)	$\lvert 7.2 - 8\rvert$
H	Adam trajectory invariant to a loss scale of 37 (five steps)	2.6×10^{-9}
J	Trace increment (E3) (relative)	1.2×10^{-13}
K1	Largest increase of $F = J\,\mathrm{tr}\,\Sigma$ along the path	-7.8×10^{-9}
K2	Final F divided by its lower bound $(\mathrm{tr}\,\hat{S}^{1/2})^2/T$	1.00028
K3	Relative distance of $\Sigma/\,\mathrm{tr}\,\Sigma$ from $\hat{S}^{1/2}/\,\mathrm{tr}\,\hat{S}^{1/2}$ at the end	1.3×10^{-2}

VII. REPLAY OF THE RELEASED META PHASE

The script `resim_meta.py` replays the released meta phase without any forecaster: plain SGD on P from $P_0 = I$, with $L = \mathrm{tril}(P) + \varepsilon I$, the released meta learning rate and 300 meta steps, the learning rate halved every 100 steps, and the constant moment $\kappa S_{0\mathrm{n}}$ in place of the query moments $\hat{S}_t$. Here $S_{0\mathrm{n}}$ is the normalized residual moment of the untrained forecaster of the same run. Because the trajectory depends on κ only through $\eta_t\kappa$, the single scalar κ sets how far along the path the replay stops. It is fitted by minimizing the shape error

$$E(\Sigma_{\mathrm{model}}) = \frac{\|\tilde{\Sigma} - \tilde{\Sigma}_{\mathrm{model}}\|_F}{\|\tilde{\Sigma} - I\|_F}, \qquad \tilde{\Sigma} = \frac{T\,\Sigma}{\mathrm{tr}\,\Sigma}, \tag{E10}$$

between the learned Σ (from the saved `A.pth` of each QDF run) and the model. The first-order formula $I + c(S_{0\mathrm{n}} + \mathrm{Diag}\,S_{0\mathrm{n}})$ and the square-root family $(I + cS_{0\mathrm{n}})^{1/2}$ get one fitted scalar each. Table E2 lists the ten ETTh runs with $T = 96$ and one run with $T = 720$, replayed on four CPU threads. Table E3 extends the comparison to the 49 other learned matrices that were still on the server disk: all TQNet runs with $T = 720$ of the earlier protocols and all iTransformer and DLinear runs. These were replayed on the GPU of the v8 server (script `extras_v8.py`, post hoc and descriptive), with $S_{0\mathrm{n}}$ taken from the injected target of the run $I + S_{0\mathrm{n}}$ with the same seed. For these runs $\mathrm{tr}\,S_0/T$ was not recorded, so the fitted κ is not reported.

The spectral norm of the fitted first-order term, between 4.6 and 13.9 on ETTh, shows that the released budgets are far outside the regime $\alpha s \ll 1$ of Proposition 2. The replay explains the learned shape best in every one of the 59 matrices, which supports the claim that Σ is a deterministic function of the residual moment and the step budget. It still fits one scalar: on ETTh the fitted κ is 1.0 to 2.4 times the evaluation-mode energy $\mathrm{tr}\,S_0/T$, so the meta-phase residuals, which come from adapted copies with dropout active, are not predicted in scale by S_0. The replay also ignores the variation of $\hat{S}_t$ across steps, which is the likely reason for the larger errors on ETTh with $T = 720$. The diagonal-to-off-diagonal slope ratio of $\Sigma - I$ on $S_{0\mathrm{n}}$ was 2.14–3.03 in the 49 additional runs. Without any fit, Table E9 compares the replay Σ_{rep} of protocol v8 with the learned matrix of the same seed.

VIII. META STEP BUDGETS OF THE RELEASED SCRIPTS

The meta learning rate halves every 100 meta steps, so $\eta_{\mathrm{tot}} = \sum_t \eta_t$ equals 175, 193.75 and 198.44 times the meta learning rate for 300, 500 and 700 steps. Table E4 lists $\alpha = 2\eta_{\mathrm{tot}}/T$ for the settings of the letter, computed from the per-horizon values in the released dataset scripts. The scale-free intensity $\rho = \alpha\,\mathrm{tr}\,\bar{S}/T$ also needs the residual energy per horizon step under the meta-phase conditions, which we did not measure directly; the replay of Section VII estimates it up to the factor κ. Where α is small, (4) predicts $\Sigma \approx I$; on ETTh1 with $T = 720$, $\alpha = 0.005$ and the learned Σ stayed within 0.03 of I in $\|\Sigma - I\|_F/\sqrt{T}$.

The iTransformer and DLinear runs copied these meta settings from the TQNet scripts.

TABLE E2
REPLAY OF THE META PHASE: SHAPE ERRORS WITH ONE FITTED SCALAR EACH

Data	T	Seed	$\|\tilde{\Sigma} - I\|_F/\sqrt{T}$	$\|$first-order term$\|_2$	E first order	E square root	E replay	$\kappa/(\mathrm{tr}\, S_0/T)$
ETTh1	96	2021	0.443	4.6	0.166	0.109	0.075	1.00
ETTh1	96	2022	0.443	4.7	0.170	0.106	0.074	1.00
ETTh1	96	2023	0.442	4.6	0.167	0.110	0.078	1.00
ETTh1	96	2024	0.444	4.7	0.165	0.109	0.077	1.00
ETTh1	96	2025	0.441	4.7	0.168	0.111	0.077	1.00
ETTh2	96	2021	0.803	10.8	0.234	0.149	0.115	2.39
ETTh2	96	2022	0.810	10.9	0.235	0.149	0.116	2.44
ETTh2	96	2023	0.801	10.7	0.236	0.149	0.117	2.42
ETTh2	96	2024	0.810	10.9	0.233	0.148	0.116	2.43
ETTh2	96	2025	0.811	10.9	0.239	0.146	0.113	2.41
ETTh2	720	2021	0.508	13.9	0.247	0.237	0.202	1.86

TABLE E3
SHAPE ERRORS OF THE REMAINING 49 LEARNED MATRICES (RANGE OVER SEEDS, ONE FITTED SCALAR EACH)

Backbone	Data	T	Runs	$\|\tilde{\Sigma} - I\|_F/\sqrt{T}$	E first order	E square root	E replay
TQ	ETTh1	720	5	0.028	0.187–0.192	0.158–0.161	0.140–0.143
TQ	ETTh2	720	5	0.506–0.510	0.245–0.250	0.236–0.238	0.200–0.203
TQ	ETTm1	720	3	0.283–0.284	0.237–0.240	0.112–0.115	0.083–0.085
TQ	ETTm2	720	3	0.249	0.283–0.284	0.134–0.135	0.118
TQ	Weather	720	3	0.310–0.311	0.257–0.259	0.097–0.098	0.053–0.054
DL	ECL	96	5	0.472	0.166–0.167	0.089–0.090	0.016
DL	Weather	96	5	0.467	0.185–0.188	0.051–0.064	0.029–0.046
DL	Weather	720	5	0.310–0.311	0.261	0.093	0.048
iT	ECL	96	5	0.523–0.529	0.188–0.192	0.105–0.110	0.041–0.045
iT	Weather	96	5	0.442–0.468	0.184–0.203	0.079–0.085	0.052–0.061
iT	Weather	720	5	0.316–0.318	0.252–0.256	0.116–0.118	0.072–0.074

TQ: TQNet; DL: DLinear; iT: iTransformer. DLinear and iTransformer used the meta settings of the TQNet scripts.

IX. ALL SEVENTEEN SETTINGS OF THE EARLIER PROTOCOLS

Table E5 lists QDF and the validated surrogate $I + aS_{0\mathrm{n}}$ in all 17 settings of protocols v3–v5 and v7, with ETTh on CPU and all other settings on GPU. QDF used its released meta settings. Averaged over the 17 settings, the surrogate was -0.03 percentage points below QDF (Wilcoxon $p = 0.64$ over settings).

X. PROTOCOL V8: VALIDATED QDF AND TWO SURROGATES ON ONE GPU

Design. Protocol v8 (`protocol_v8.json`, SHA-256 prefix `2f951e7b203705e0`) was designed after a mock review of an earlier draft and frozen before its runs, so it is post hoc with respect to all earlier results. It repeats the eight analysis settings on one rented server (NVIDIA RTX 4080 SUPER, PyTorch 2.13.0 with CUDA 13.0, QDF commit `eb0693a9`, data files identified by SHA-256). Per seed it runs the identity control; QDF with its meta learning rate scaled by $m \in \{\frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8\}$ (TQNet) or $m \in \{\frac{1}{4}, \frac{1}{2}, 1, 2, 4\}$ (iTransformer), grids as large as those of a; $I + aS_{0\mathrm{n}}$ and $I + \frac{a}{2}(S_{0\mathrm{n}} + \mathrm{Diag}\, S_{0\mathrm{n}})$ over the grid of a of the letter; the replay with constant moment $k\,(\mathrm{tr}\, S_0/T)S_{0\mathrm{n}} = kS_0$ for $k \in \{\frac{1}{2}, 1, 2\}$; and, at seed 2021, an exact repeat of the identity control. The doubled-diagonal form is scaled by $\frac{1}{2}$ so that its trace equals that of $I + aS_{0\mathrm{n}}$. The replay runs the released meta updates of P (same meta learning rate, step count and halving schedule, $P_0 = I$, $L = \mathrm{tril}(P) + 10^{-6}I$) on the GPU and injects the result; its code equals `resim_meta.py` to 1.6×10^{-16} on a random moment. Every validated arm takes, per seed, the grid value with the lowest best validation MSE; ties go to $m = 1$ for QDF, then to the smaller value. Failed runs are ineligible and retried once. With $n = 8$ settings, a verdict needs its condition in $k = 6$ settings,

TABLE E4
RELEASED META SETTINGS AND CUMULATIVE STEP SIZES

Dataset	T	Meta learning rate	Meta steps	η_{tot}	$\alpha = 2\eta_{\text{tot}}/T$
ETTh1	96 / 720	0.2 / 0.01	300	35.0 / 1.75	0.729 / 0.0049
ETTh2	96 / 720	0.2 / 0.2	300	35.0 / 35.0	0.729 / 0.097
ETTm1	96 / 720	0.1 / 0.2	500	19.4 / 38.8	0.404 / 0.108
ETTm2	96 / 720	0.2 / 0.2	500	38.8 / 38.8	0.807 / 0.108
Weather	96 / 720	0.1 / 0.2	700	19.8 / 39.7	0.413 / 0.110
ECL	96	0.1	300	17.5	0.365

TABLE E5
QDF AND THE VALIDATED SURROGATE IN ALL 17 SETTINGS OF THE EARLIER PROTOCOLS

	Data–T	MSE_I	$\bar{g}_{\text{QDF}}$ (%)	$\bar{g}_{\text{val}}$ (%)	$\bar{g}_{\text{val}} - \bar{g}_{\text{QDF}}$
TQ	ETTh1-96*	0.378	1.95 ± 1.20 (5)	1.92 ± 1.23 (5)	-0.03 ± 1.96
	ETTh1-720	0.496	1.49 ± 3.32 (3)	0.84 ± 3.49 (2)	-0.65 ± 4.56
	ETTh2-96*	0.293	2.27 ± 0.95 (5)	2.36 ± 0.39 (5)	0.08 ± 1.28
	ETTh2-720*	0.435	3.46 ± 1.08 (5)	2.31 ± 1.53 (5)	-1.15 ± 2.40
	ETTm1-96*	0.310	0.75 ± 1.02 (3)	0.54 ± 0.80 (3)	-0.21 ± 0.40
	ETTm1-720	0.449	0.54 ± 4.90 (2)	1.15 ± 0.71 (3)	0.61 ± 4.20
	ETTm2-96*	0.172	0.89 ± 0.43 (3)	0.37 ± 0.44 (3)	-0.52 ± 0.20
	ETTm2-720*	0.391	1.03 ± 0.46 (3)	0.83 ± 0.59 (3)	-0.19 ± 0.36
	Wth-96	0.160	0.01 ± 2.91 (1)	0.03 ± 1.66 (1)	0.01 ± 1.57
	Wth-720*	0.344	0.51 ± 0.21 (3)	0.49 ± 0.77 (3)	-0.02 ± 0.70
	ECL-96	0.135	0.44 ± 0.24 (3)	0.28 ± 0.51 (3)	-0.15 ± 0.39
iT	ECL-96*	0.150	0.91 ± 0.38 (5)	0.76 ± 0.26 (5)	-0.15 ± 0.36
	Wth-96	0.176	-1.76 ± 0.90 (0)	-0.60 ± 0.70 (0)	1.16 ± 1.05
	Wth-720	0.359	-0.53 ± 0.41 (1)	-0.03 ± 0.44 (2)	0.50 ± 0.34
DL	ECL-96	0.195	0.01 ± 0.01 (5)	0.01 ± 0.00 (5)	0.00 ± 0.01
	Wth-96	0.196	-0.29 ± 0.70 (1)	-0.04 ± 0.01 (0)	0.25 ± 0.69
	Wth-720	0.347	0.11 ± 0.45 (2)	0.14 ± 0.39 (3)	0.02 ± 0.07

TQ: TQNet; iT: iTransformer; DL: DLinear; Wth: Weather. Mean $\pm$ half-width of the 95% t-interval over seeds (five, or three for TQ on ETTm, Wth and ECL); seeds with positive gain in parentheses; the difference is paired by seed. MSE_I: test MSE of the identity control. *: analysis setting.

with a positive gain in at least four of five or three of three seeds; R is relative to validated QDF (primary) and to released QDF (secondary).

Results. All 778 runs succeeded (778 records with a finite test and validation MSE). Table E6 lists every arm with its 95% interval. The frozen verdicts are in Table E12. Against validated QDF, $I + aS_{0\text{n}}$ reached $R \geq 0.8$ with enough positive seeds in 6 settings, $I + \frac{a}{2}(S_{0\text{n}} + \text{Diag}\, S_{0\text{n}})$ in 4, the replay with $k = 1$ in 6 and the replay with validated k in 5. Against released QDF the counts were 5 for $I + aS_{0\text{n}}$ and 5 for the replay with $k = 1$. Over settings, $I + aS_{0\text{n}}$ was on average -0.32 points below validated QDF (lower in 6 of 8 settings, Wilcoxon $p = 0.039$) and -0.32 points below released QDF (lower in 8, $p = 0.008$). For the replay with $k = 1$ the differences were -0.40 ($p = 0.312$) and -0.40 points (lower in 7, $p = 0.023$). Validated QDF differed from released QDF by 0.00 points on average ($p = 0.20$) and chose $m = 1$ in 12 of 32 seeds. The doubled diagonal was better than the plain form on average in 3 of 8 settings ($p = 0.64$).

Dose responses. Tables E7 and E8 list the mean gain for every grid value. QDF's mean gain was negative in 1 of 54 combinations of setting and m (minimum -0.05%); $I + aS_{0\text{n}}$ was negative in 12 of 54 (minimum -4.51%). On ETTh the best single m on test data exceeded 1, so the released budget was below the best one there.

Replay without a fit. Table E9 compares the replay Σ_{rep} ($k = 1$) with the learned Σ of released QDF at the same seed. Where the shape error is large, on ETTh2 at both horizons and on ETTm2 with $T = 96$, the replay moves less far from I than the learned matrix, which is consistent with the fitted κ above 1 on ETTh2 (Table E2); on ETTm2 with $T = 720$ it moves further.

Cost. QDF's meta phase took 36–75% of the wall-clock time of a released QDF run (87–593 s, with concurrent jobs on one GPU). Computing S_0 took at most 26.8 s and the replay 0.08–10.5 s. All surrogate runs still executed the inert meta phase of the identity control, so that they share its random-number consumption and batch order.

Platforms. The exact repeats of the identity control reproduced the test MSE exactly in all eight settings. The 34 runs of QDF and the identity control that v8 shares with protocols v5 and v7, which ran on a second server of the same type, had identical test MSE. On ETTh, the 30 shared runs differed from the CPU runs of v3 and v4 by at most 2.59%.

Engineering events. All are recorded in `protocol_v8_erratum.json` and none changed arms, settings, seeds or rules. (i) The first smoke test failed at import because a library was missing on the new server; it was installed in the version used before, and smoke runs are excluded from all analyses. (ii) Three minutes after the first launch, before any run had finished, the runner was stopped and given a GPU-memory budget to avoid out-of-memory failures; the partial work directories were set aside. (iii) The frozen file gives a freezing time 15 minutes after the actual launch; this was a typing error, and the file was left unchanged to keep its hash valid. (iv) After 661 successful runs the data disk filled with event logs that the unmodified QDF code writes, one run (the exact repeat on ETTh2 with $T = 720$) failed, and the host shut down; after the logs were deleted, the runner resumed, skipped recorded runs and retried the failed one, which succeeded.

TABLE E6
PROTOCOL V8: MEAN GAIN OVER THE IDENTITY CONTROL (%) WITH 95% INTERVAL AND POSITIVE SEEDS

Data–T	MSE_I	QDF released	QDF validated	$I + aS_{0n}$	$I + \frac{a}{2}(S_{0n} + \mathrm{Diag}\, S_{0n})$	replay, $k = 1$	replay, validated k
ETTh1-96	0.378	2.02 ± 1.14 (5)	1.96 ± 0.76 (5)	1.66 ± 1.20 (5)	2.09 ± 1.11 (5)	1.96 ± 0.83 (5)	1.98 ± 0.68 (5)
ETTh2-96	0.293	2.56 ± 0.83 (5)	3.44 ± 1.44 (5)	2.33 ± 1.25 (5)	2.50 ± 0.34 (5)	1.64 ± 1.53 (5)	1.89 ± 1.16 (5)
ETTh2-720	0.433	3.19 ± 1.46 (5)	2.79 ± 1.83 (5)	2.28 ± 2.06 (5)	1.12 ± 0.76 (5)	1.34 ± 0.73 (5)	1.56 ± 1.24 (5)
ETTm1-96	0.310	0.75 ± 1.02 (3)	0.51 ± 0.97 (3)	0.54 ± 0.80 (3)	0.50 ± 0.75 (3)	0.75 ± 1.07 (3)	0.20 ± 1.95 (2)
ETTm2-96	0.172	0.89 ± 0.43 (3)	0.87 ± 0.67 (3)	0.37 ± 0.44 (3)	0.35 ± 0.39 (3)	0.72 ± 0.26 (3)	0.81 ± 0.39 (3)
ETTm2-720	0.391	1.03 ± 0.46 (3)	0.99 ± 0.54 (3)	0.83 ± 0.59 (3)	0.58 ± 0.97 (3)	0.95 ± 0.35 (3)	0.89 ± 0.39 (3)
Wth-720	0.344	0.51 ± 0.21 (3)	0.40 ± 0.56 (3)	0.49 ± 0.77 (3)	0.40 ± 0.46 (3)	0.36 ± 0.48 (3)	0.41 ± 0.33 (3)
iT ECL-96	0.150	0.91 ± 0.38 (5)	0.89 ± 0.35 (5)	0.76 ± 0.26 (5)	0.80 ± 0.25 (5)	0.94 ± 0.36 (5)	0.89 ± 0.35 (5)

TQNet unless marked iT (iTransformer); Wth: Weather. Five seeds on ETTh and iT ECL, three otherwise. a, m and k validated per seed.

TABLE E7
PROTOCOL V8: MEAN GAIN (%) OF QDF AGAINST THE META-LEARNING-RATE MULTIPLIER m AND OF THE REPLAY AGAINST k

	QDF, m							replay, k		
Data–T	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8	$\frac{1}{2}$	1	2
ETTh1-96	0.40	0.97	1.60	2.02	2.61	2.67	2.52	1.82	1.96	2.47
ETTh2-96	1.20	1.45	1.86	2.56	3.04	3.38	3.57	1.42	1.64	2.02
ETTh2-720	0.22	0.72	1.53	3.19	4.51	4.56	4.37	0.79	1.34	2.25
ETTm1-96	0.50	0.13	0.16	0.75	−0.05	0.42	0.28	0.14	0.75	−0.09
ETTm2-96	0.54	0.66	0.79	0.89	0.90	0.91	0.87	0.60	0.72	0.81
ETTm2-720	0.77	0.90	0.99	1.03	1.02	0.96	1.05	0.95	0.95	0.89
Wth-720	0.38	0.19	0.26	0.51	0.41	0.40	0.28	0.28	0.36	0.48
iT ECL-96	–	0.75	0.78	0.91	0.88	0.69	–	0.77	0.94	0.92

XI. PROTOCOL V9: FRESH SEEDS

Design. Protocol v9 (`protocol_v9.json`, SHA-256 prefix `926f4516311d8c09`) repeats every arm, grid, selection rule and decision rule of v8 with seeds 2026–2028 in all eight analysis settings. No earlier protocol used these seeds, so neither the settings nor the grids were chosen on them. It ran on a second server of the same type, in an isolated environment with the library versions of v8; before the grid, 16 runs of v8 (identity control and released QDF at seed 2021 in every setting) were repeated and had identical test and validation MSE. The erratum file

TABLE E8
PROTOCOL V8: MEAN GAIN (%) OF $I + aS_{0n}$ AGAINST a

Data–T	0.01	0.03	0.1	0.25	0.3	0.5	1	2
ETTh1-96	0.17	0.33	1.22	2.60	–	2.54	2.41	2.28
ETTh2-96	0.31	0.71	1.39	3.17	–	3.32	2.49	0.51
ETTh2-720	0.42	3.15	3.99	3.29	–	2.88	2.42	1.47
ETTm1-96	0.13	0.49	0.23	0.68	–	0.71	0.59	0.23
ETTm2-96	0.21	0.45	0.80	0.92	–	0.37	−0.81	−2.88
ETTm2-720	0.88	0.63	−1.07	−2.59	–	−3.19	−3.74	−4.51
Wth-720	0.29	0.51	−0.21	−1.02	–	−1.61	−2.50	−3.27
iT ECL-96	0.18	0.46	0.73	–	0.75	–	0.66	–

The grid was $\{0.01, 0.03, 0.1, 0.25, 0.5, 1, 2\}$ for TQNet and $\{0.01, 0.03, 0.1, 0.3, 1\}$ for iTransformer.

TABLE E9
PROTOCOL V8: REPLAY WITHOUT A FIT AGAINST THE LEARNED Σ OF THE SAME SEED

Data–T	Seeds	$\|\tilde{\Sigma} - I\|_F/\sqrt{T}$ learned	$\|\tilde{\Sigma} - I\|_F/\sqrt{T}$ replay	E replay	R_{rep}
ETTh1-96	5	0.442–0.447	0.438–0.440	0.076–0.082	1.00
ETTh2-96	5	0.801–0.814	0.556–0.565	0.316–0.318	0.48
ETTh2-720	5	0.507–0.511	0.348–0.349	0.348–0.353	0.48
ETTm1-96	3	0.374–0.383	0.373–0.383	0.032–0.035	1.46
ETTm2-96	3	0.697–0.707	0.476–0.481	0.326–0.335	0.83
ETTm2-720	3	0.249	0.316–0.317	0.302	0.96
Wth-720	3	0.310–0.311	0.316–0.318	0.057–0.059	0.92
iT ECL-96	5	0.523–0.529	0.539–0.546	0.053–0.057	1.05

Learned Σ from released QDF; ranges over seeds. R_{rep}: recovery ratio of the replay against validated QDF.

(`fc01300ac69d8dd1`) corrects the recorded freezing time, which was estimated when the file was written; the order of engineering check, freeze and first run is unchanged.

Results. All 590 runs succeeded. Table E10 lists every arm. Released QDF's mean gain was positive in all 8 settings and in every seed in 7; 6 settings passed the entry rule of the analysis. Against validated QDF, $I + aS_{0n}$ reached $R \geq 0.8$ with all seeds positive in 3 settings and the replay with $k = 1$ in 4 (verdicts MIXED and PARTIAL); the doubled diagonal again made no consistent difference (NO_DIFFERENCE). The mean differences from validated QDF were -0.17 points ($p = 0.46$) and -0.50 points ($p = 0.078$). On ETTh2 with $T = 96$ validation chose $m = 8$ in every seed.

Post hoc summaries. Neither was frozen. (i) Choosing, per seed, whichever of $I + aS_{0n}$ and the replay has the lower best validation MSE reached $R \geq 0.8$ with all seeds positive in 5 settings, -0.09 points from validated QDF on average. (ii) Pooling the seeds of v8 and v9 (eight on ETTh and for iTransformer, six otherwise) and requiring a positive gain in all seeds but one, $I + aS_{0n}$ recovered at least 80% in 5 settings, the replay in 6 and the validation choice in 6, with mean differences of -0.27 and -0.44 points. (iii) Relative to released QDF, with all seeds positive, $I + aS_{0n}$ reached $R \geq 0.8$ in 4 settings and the replay in 5. (iv) Restricted to the 6 settings in which released QDF passed the entry rule on the v9 seeds (all but ETTm1, with one negative seed, and Weather, at 0.37%), the frozen counts become 2 and 3. (v) Summed over the eight settings, the gains of $I + aS_{0n}$ and the replay were 78% and 73% of those of validated QDF in v8 and 90% and 70% in v9.

XII. FROZEN VERDICTS

Each protocol fixed its analysis settings, decision rules and thresholds before its own runs. Table E12 reports every verdict as the scripts produced it. The v3 verdict was produced under a recorded amendment, frozen before the remaining runs, that moved all $T = 96$ runs of v3 to the CPU of a rented server (the GPU was not used), so that every number in a setting comes from one platform; the 28 runs repeated on both platforms differ by at most 0.40% in test MSE. With n analysis settings, a verdict requires its condition in at least $k = \lceil 2n/3 \rceil$ of them.

TABLE E10
PROTOCOL V9: MEAN GAIN OVER THE IDENTITY CONTROL (%) WITH 95% INTERVAL AND POSITIVE SEEDS

Data–T	QDF released	QDF validated	$I + aS_{0n}$	$I + \frac{a}{2}(S_{0n} + \text{Diag}\, S_{0n})$	replay, $k = 1$	selected m
ETTh1-96	1.73 ± 1.03 (3)	2.75 ± 1.05 (3)	3.01 ± 0.47 (3)	1.93 ± 2.07 (3)	1.89 ± 0.21 (3)	4/2/2
ETTh2-96	1.30 ± 1.62 (3)	2.60 ± 1.09 (3)	1.65 ± 1.37 (3)	1.95 ± 0.99 (3)	0.77 ± 1.95 (2)	8/8/8
ETTh2-720	3.56 ± 0.66 (3)	2.68 ± 1.48 (3)	3.29 ± 2.50 (3)	1.59 ± 3.08 (3)	1.63 ± 2.86 (3)	1/0.5/0.5
ETTm1-96	1.15 ± 3.71 (2)	1.12 ± 4.04 (2)	0.92 ± 3.21 (2)	0.90 ± 3.29 (3)	1.11 ± 3.68 (2)	1/8/0.25
ETTm2-96	1.36 ± 2.11 (3)	1.39 ± 2.64 (3)	0.83 ± 2.80 (3)	1.21 ± 2.26 (3)	1.16 ± 1.89 (3)	8/8/8
ETTm2-720	1.26 ± 1.10 (3)	1.38 ± 1.00 (3)	0.98 ± 1.19 (3)	1.04 ± 0.60 (3)	1.24 ± 0.93 (3)	8/4/8
Wth-720	0.37 ± 0.17 (3)	0.41 ± 0.84 (3)	0.50 ± 0.67 (3)	0.30 ± 0.77 (3)	0.55 ± 0.26 (3)	0.25/2/0.5
iT ECL-96	0.66 ± 0.40 (3)	0.70 ± 0.72 (3)	0.49 ± 0.35 (3)	0.67 ± 0.30 (3)	0.73 ± 0.27 (3)	4/4/0.5

TQNet unless marked iT (iTransformer); Wth: Weather. Seeds 2026–2028. a and m validated per seed.

TABLE E11
PER-SETTING RECOVERY RATIOS OF THE CONTROLS

	$I + \frac{a}{2}(S_{0n} + \text{Diag}\, S_{0n})$		earlier protocols	
Data–T	v8	v9	R_{C_y}	R_{rot}
ETTh1-96	1.06	0.70	0.50	0.10
ETTh2-96	0.73	0.75	0.49	−0.01
ETTh2-720	0.40	0.59	0.39	0.00
ETTm1-96	0.97	0.80	0.26	−0.18
ETTm2-96	0.40	0.87	−0.77	−0.09
ETTm2-720	0.59	0.75	0.43	−0.01
Wth-720	1.01	0.75	0.08	−0.25
iT ECL-96	0.90	0.95	0.84	−0.14

The doubled diagonal with validated a, relative to validated QDF of the same protocol. Earlier protocols, relative to released QDF, ETTh on CPU: R_{C_y}, label covariance in place of S_{0n}; R_{rot}, spectrum of Σ_a with Haar-random eigenvectors.

On ETTm and Weather ($n = 4$, $k = 3$), the surrogate reached $R_{\text{val}} \geq 0.8$ with all seeds positive in two settings, hence MIXED. For iTransformer and DLinear only one setting qualified ($n = 1 < 2$). In v1, a setting showed a Σ effect if QDF's mean gain over the identity control reached half the gain implied by the published QDF and TQNet numbers. On ETTh1 with $T = 720$ the measured gain was 1.8%, less than a quarter of the published 8.1%, so the setting was classed NO_SIGMA_EFFECT and the overall verdict was INCONCLUSIVE. The three other settings became the ETTh analysis settings of all later protocols.

The secondary arms of v2 replaced the learned Σ by simpler matrices of the same trace. On the three ETTh analysis settings, the top eight eigenpairs of the learned Σ recovered $R = 1.17$–1.42, its diagonal at most 0.28, and its mean eigenvalue between -0.09 and 0.00.

TABLE E12
VERDICTS OF THE FROZEN PROTOCOLS

Protocol (settings)	Question	Verdict
v1 (ETTh, TQNet, CPU)	QDF beats the identity control, per setting	SIGMA_EFFECT in 3 of 4
	Overall	INCONCLUSIVE
v2 (ETTh, TQNet, CPU)	Trace-matched $I + cS_0$ recovers the gain	PLUGIN_SUFFICES
v3 (ETTh, TQNet, CPU)	Validated $I + aS_{0\mathrm{n}}$ recovers the gain	META_LEARNING_UNNECESSARY
v4 (ETTh, TQNet, CPU)	Residual moment vs. label covariance	RESIDUAL_BETTER
	Eigenvectors vs. spectrum only	STRUCTURE_MATTERS
	Ledoit–Wolf intensity	AUTO_SHRINK_FAILS
v5 (ETTm, Weather, ECL; TQNet, GPU)	Validated $I + aS_{0\mathrm{n}}$ recovers the gain	MIXED
	Residual moment vs. label covariance	RESIDUAL_BETTER
	Eigenvectors vs. spectrum only	STRUCTURE_MATTERS
v6 (four v5 settings, GPU)	Re-injected learned Σ reproduces QDF	PIPELINE_OK
	Trace-matched $I + cS_0$ recovers the gain	PARTIAL
	Trace-matched $I + c(S_0 + \operatorname{Diag} S_0)$ recovers the gain	RECOVERS
v7 (ECL, Weather; iTransformer, DLinear, GPU)	All questions	INSUFFICIENT_CELLS
v8 (eight analysis settings, GPU)	Validated $I + aS_{0\mathrm{n}}$ recovers the gain of validated QDF	SURROGATE_MATCHES_TUNED_QDF
	Validated $I + \frac{a}{2}(S_{0\mathrm{n}} + \operatorname{Diag} S_{0\mathrm{n}})$ beats validated $I + aS_{0\mathrm{n}}$	NO_DIFFERENCE
	Replay with $k = 1$ recovers the gain of validated QDF	REPLAY_RECOVERS
	Replay with validated k recovers the gain of validated QDF	PARTIAL
v9 (as v8, seeds 2026–2028, GPU)	Validated $I + aS_{0\mathrm{n}}$ recovers the gain of validated QDF	MIXED
	Validated $I + \frac{a}{2}(S_{0\mathrm{n}} + \operatorname{Diag} S_{0\mathrm{n}})$ beats validated $I + aS_{0\mathrm{n}}$	NO_DIFFERENCE
	Replay with $k = 1$ recovers the gain of validated QDF	PARTIAL